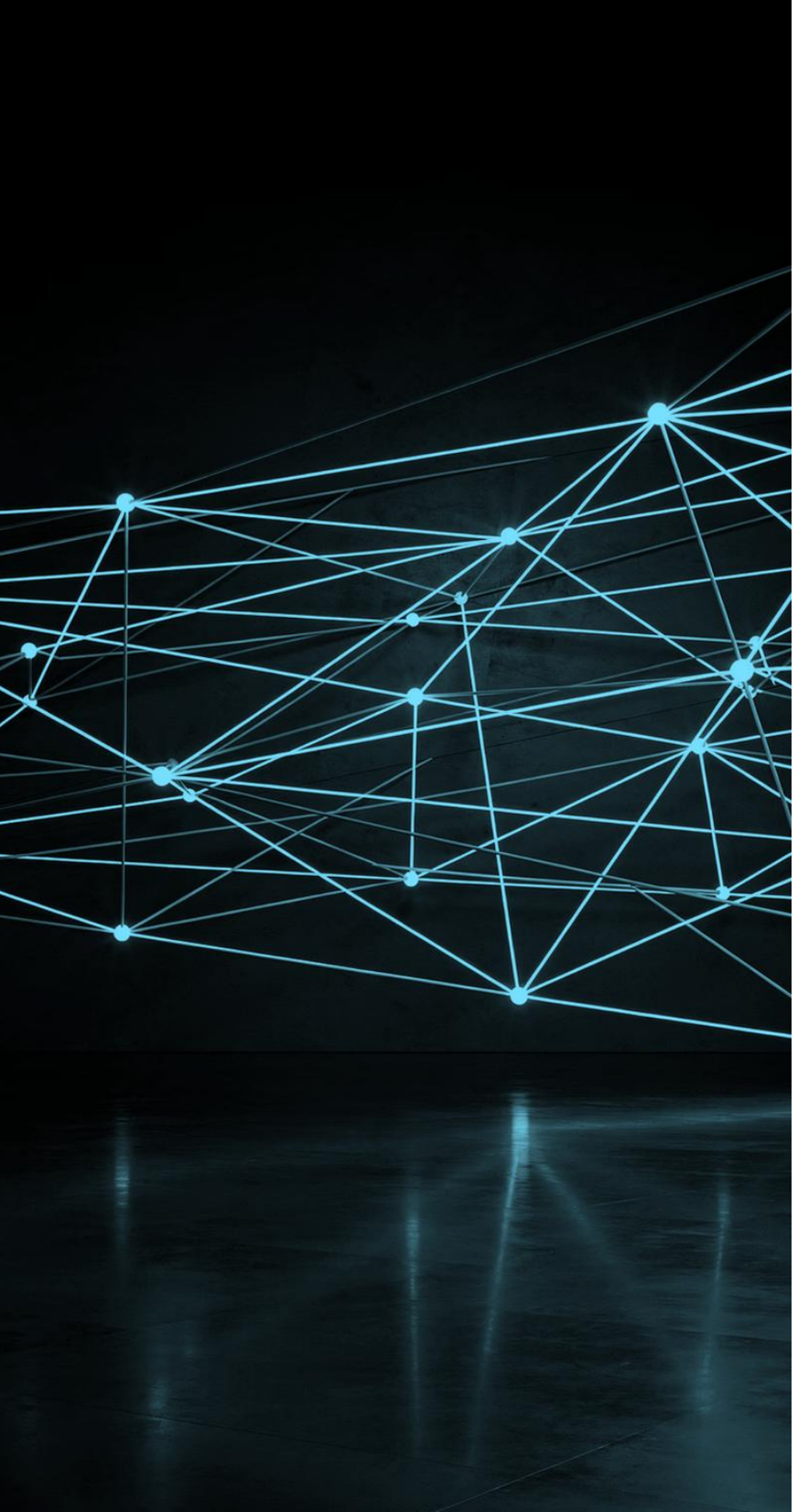


Ambient Systems

Integrated technical expertise in engineering, FP&A, AI, and IP.

Turning five years of deep-tech R&D into specialized consulting that delivers direct results.

Based from Belgrade with global operations in Hong Kong.

Introduction

Agency Overview

Driving financial performance through engineering, AI, and IP.

Background

Offering expertise after five years of building an award-winning, PCT-patented NUS deep-tech AI technology startup that achieved technical readiness (TRL 6).

Our Mandate

- ✓ Industrial applications
- ✓ Streamlined FP&A
- ✓ Intellectual Property R&D
- ✓ Software subscription (in development)

Service Lines:

1. **Civil Engineering & Energy Systems**
   Drafting and load calculations.
   Solar array design, microgrid optimization, building energy efficiency.
2. **Supply-Chain Management & Vendor Optimization**
   Sourcing strategy, customs optimization, vendor scorecards.
3. **Vertical Farming / CEA Optimization**
   Layout design, yield modeling, environmental controls.
4. **IP Strategy & Patent Support**
   Prior art searches, patent filing support, freedom-to-operate analysis.

Active on Leading Platforms: Upwork, Freelancer.com, Catalant, AlphaSights, and —connecting with clients who need immediate, specialized expertise.

Case Study

End-to-End Climate Management Solution for Indoor Farming

SITUATION or BUSINESS CHALLENGE

Indoor farmers face significant crop loss and high operational costs due to the inability to maintain optimal climate conditions. Existing solutions are fragmented, manual, and labor-intensive, often leading to unmonitored temperature and humidity imbalances that promote mold and pests, ultimately reducing yield quality and quantity.

APPROACH

- Defined the **full-stack ML solution** (sensors, optimization engine, dashboard) and the scalable B2B SaaS + Hardware business model.
- Secured pilot **Letters of Intent (LOIs)** with two key vertical farm operators to validate demand and establish the initial pipeline.
- Developed the financial model projecting a **24% improvement in client margins** with a rapid **4–6-month customer payback period**.
- Produced all investment deliverables (Investment Deck, Business Plan) and defined the IP/patent strategy for the core optimization algorithm.

OUTCOMES ACHIEVED

- 24% ROI Model: Developed a model projecting a 24% improvement in client margins with a rapid 4–6-month customer payback.
- Validation: Achieved critical market validation and established a pipeline through secured commercial LOIs.
- IP Foundation: Defined the patent strategy for the core optimization algorithm.

Previous Experience

Results achieved personally at prior ventures.

Case studies in distributed mining, SEC big pharma disclosures, and image recognition.

	Distributed Mining Company	Conflict Minerals	Image Neural Network
Challenge	Uncontrolled OpEx (Energy Costs). No link between real-time operational data and financial planning; resulting in reactive procurement.	Urgent SEC 3TG Conflict Minerals disclosure. Major pharmaceutical client required rapid compliance under a tight regulatory deadline.	Startup needed a robust, high-accuracy AI vision model to automatically detect artisanal mining by analyzing satellite imagery across complex biomes.
Action	Developed Big Data predictive OpEx models using operational and market data. Integrated models for real-time cost visibility at site level.	Led data analysis and consolidated raw data from 20+ international offices. Advised on compliance risk mitigation for SEC disclosure requirements.	Led as Principal Investigator (PI). Designed and implemented Convolutional Neural Network (CNN) models. Developed an automated data collection and prediction pipeline.
Result	12% sustained reduction in global energy costs, achieving millions in OpEx savings.	Successfully met critical U.S. regulatory disclosure. Delivery executed under urgent, one-day notice.	Delivered a validated CNN image classification model and system architecture.

Solution for PetroVietnam: Intelligent Twin (PoC)

- **Challenge #1**: Complex business processes involving multiple approvals across departments result in prolonged processing times and obscure bottlenecks
- **Collaboration Need #3**: An intelligent digital twin

(30-Day PoC Sprint)

- **Week 1: Ingest & Define**
  - Access essential PVN data (manuals, regulations, logs).
  - **Deliverable: Knowledge Status Quo**
- **Week 2: Digital Twin Build**
  - Configure the Ambient Platform to mirror the one selected process.
  - **Deliverable: Functional Process Twin**
- **Week 3: Simulate & Analyze**
  - Run AI agent scenarios (The "Policy Sandbox") to test a regulatory change.
  - **Deliverable: Impact Analysis Report**
- **Week 4: Review & Scale**
  - Present findings and demonstrated financial/efficiency gains.
  - **Deliverable: Roadmap for Adoption**

PETROVIETNAM

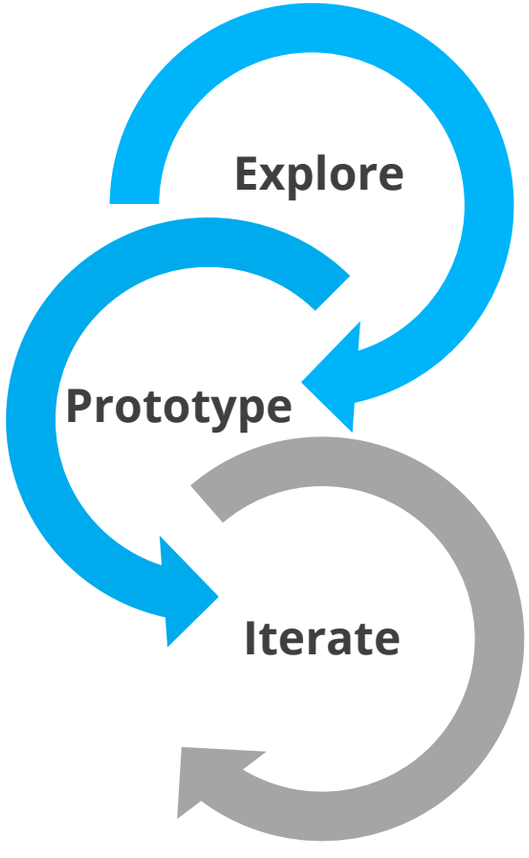
Potential Outcomes:

- **Identified Bottlenecks:** A data-driven report pinpointing exactly where "prolonged processing times" occur within the piloted workflow.
- **Sandbox Simulation Results:** A "safe test" verifying how a specific regulatory change (e.g., new safety standard) would impact operational speed before real-world implementation.
- **Unified Governance Dashboard:** A functional instance of the **Ambient Platform** (Overview Page) visualizing the process flow, "Data Status," and "Recommendations" for optimization.

Business model

Our approach

Transparent Pricing



Book a Free 30-Minute Call

- We discuss your challenge, goals, and constraints.
- No pitch–just a questioning conversation to see if we can solve your problem.

Receive Fixed-Price Proposal in 48 Hours

- Clear scope, defined deliverables, flat fee.
- You know exactly what you're paying and what you're getting before committing a dollar.

Get Results in 30-60 Days

- Fast deliverable outcomes, tangible outcomes, and clear next steps.
- Most engagements pay for themselves in the first month.

Most engagements start between **€1,500–€3,500 fixed price**.
No hourly billing. No scope creep. No surprise invoices.

Let us bring the tech to you.

Current: Based in Belgrade with global operations in Hong Kong, available for remote engagements globally.
Future: Centralizing global operations in Belgrade after mid-2026 to serve European clients.

Languages: Native Serbian & English (fluent); Intermediate French

Ready to Start?

Five years of deep-tech R&D evolved into something clients need, affordably.

First call is free.
Quote in 48 hours.
Results in 30 days.

Book Your Calendar

Ivan Damnjanovic
Founder
Technical Lead & Systems Architect

Entrepreneur and inventor from Belgrade with 10 years of experience across Asia (Singapore, Nepal, Hong Kong). Track record scaling YC-backed startup and cross-border intellectual property ventures.

Languages: Native Serbian & English (fluent); Intermediate French

Why Ambient Systems for Inno Vietnam–Japan

- Proven impact in global supply-chain and energy projects
  - 12 % OpEx reduction (decentralised mining client)
  - 24 % ROI (Singapore-based vertical farming project)
- Repeated collaboration with Japanese corporates and IP programmes
  - Panasonic, Google Cloud, NUS GRIP
- Expert team, Hong Kong incorporated, Singapore dedicated contact address
  - rapid 4-week delivery, fixed-price engagements
- Experienced in secure contracts, timely execution, and joint IP development
  - WIPO-certified processes
- Established network in Asia-Pacific and Europe
- Invoice from Hong Kong entity, Singapore mailing address provided
  - Immediately available for on-site or remote collaboration